Impact of Open-Source Software in Adopting and Implementing Electronic Medical Record Systems

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Abstract

This conceptual study focuses on the adoption of open-source Health Information Technology (HIT) in the delivery of health care. More specifically, it looks into the challenges that face the adoption and the implementation of Electronic Medical Record (EMR) systems. Open-Source software is an alternative to proprietary software. The study examines the specifics of open-source EMR systems in order to see how such systems may alleviate some of the challenges of the adoption and the implementation of traditional proprietary EMR systems.

Introduction

This study focuses on the adoption of open-source Health Information Technology (HIT) in the delivery of health care in ambulatory care clinics and family medicine practices. Electronic Medical Records (EMR) are one HIT technology that allows storing and accessing digital versions of patients medical records. It is agreed upon that the use of technology in healthcare is associated with many advantages. However, the penetration rate of many technologies in the healthcare sector is still low. There are challenges in adopting an implementing HIT in the medical practice.

Studying the implementation process is also a challenge. Much of what is known about HIT implementation and usage comes from the experience of a few large hospitals using proprietary HIT system (Faraj, 2008). The HIT systems in these settings are black-boxes. A single party posses the source code and makes the modification required by the stakeholders. In contrast, the success of open-source software and applications may extend to the health care environment.

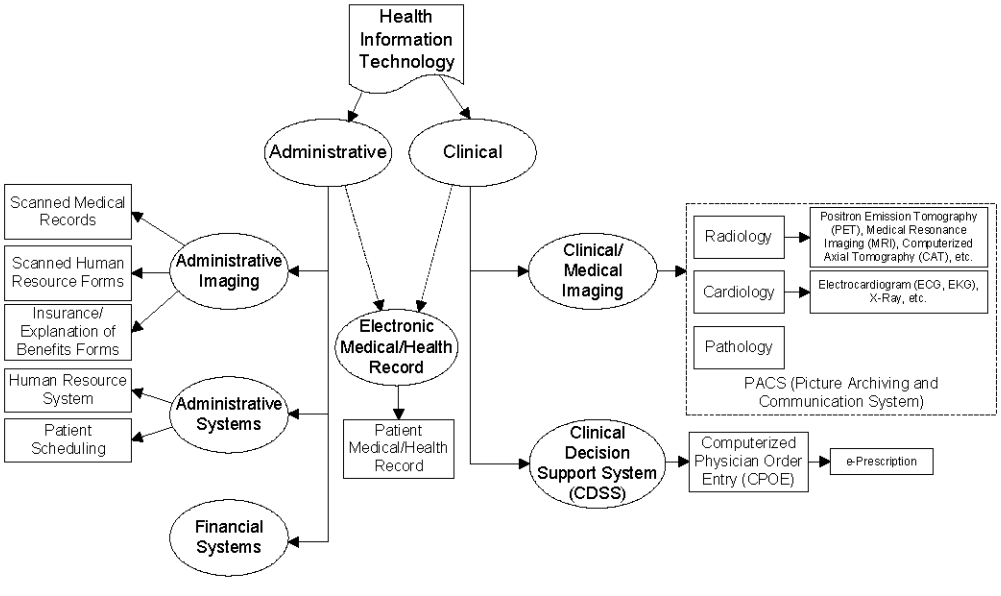
This study has two goals: first the study examines the challenges that inhibit the adoption and the implementation EMR systems in medical clinics. Second, the study looks at an alternative to traditional EMR systems to deal with the challenges. Open-Source software is software whose code is freely available and developed by different contributors. The Open-source initiative is examined as a potential alternative to proprietary EMR systems to deal with the challenges.

1. PROBLEM DEFINITION AND BOUNDARIES

The study proceeds as the following, Section 2 defines the problem and sets its boundaries. Section 3 reviews relevant literature of EMR systems classifying it within a conceptual framework. In reviewing the literature and constructing the conceptual framework, some other examples of HIT than EMR are used when the context is appropriate. In order to explain the findings of Section 2, Section 3 carries on the theoretical analysis and suggests potential theories to explain the adoption and implementation of EMR systems. Finally, a methodology for conducting the study is proposed.

1 Problem Definition and Boundaries

Health Information Technology (HIT) is a diverse set of technologies that serve in storing, managing and transmitting health information for the use of health consumers (patients), health providers (physicians, nurses, etc. . .), payers, insurers and all other groups that have an interest in health and health care (Blumenthal & Glaser, 2007).



This definition encompasses a lot of different technologies. Therefore, it is important to provide a taxonomy in order to draw the boundaries in this study. Agarwal and Angst (2006) provide a very comprehensive taxonomy of HIT. This taxonomy is depicted in Figure 1. There are two main categories of HIT systems: administrative and clinical systems. Administrative systems are those systems that are used by the administrative staff to do operational level tasks such as retrieving insurance and billing. Clinical systems on the other hand are used by medical practitioners (doctors and nurses) for core clinical tasks such as medical imaging.

Figure 1: Taxonomy of HIT, taken from (Agarwal & Angst, 2006, p. 362)

This study focuses on Electronic Medical/Health Record (EMR/EHR) systems[1]. EMR systems combines features from both administrative and clinical systems. EMRs allow storing and [1]The terms EMR and EHR are used interchangeably

accessing a digital format patient records. The digital records include both clinical and administrative information (Agarwal & Angst, 2006). Medical workers can use the EMR to carry on clinical tasks such as reviewing patients history, prescribing drugs and ordering laboratory tests, and administrative tasks such as scheduling appointments and billing.

1.1 Open-Source Software

The open-source software development movement has gained momentum in the last year and emerged as a serious competitor to the traditional proprietary software development model. The Linux operating system, the Apache web server, and the Mozilla Firefox browser are few examples of successful open-source products that gained market success over their proprietary competitors.

Although open-source development relies on voluntary participation of programmers and developers. The main development is usually done in one organization. The organization can be a community organization, commercial organization or a nonprofit organization. Different organizational models differ in their methods for funding software development, providing support and installation services, managing the code base and generating revenue (Faus & Sujansky, 2008). Indeed, several researchers looked into the economical plausibility of the open-source software model and found that the model relies on sound economical principles despite the initial impression that it offers something for nothing (Muffatto, 2006).

Open-Source Electronic Medical Records

Open-source was traditionally used by tech savvy users. This trend is however changing and we are seeing more open-source software replacing proprietary software in many areas. Several open-source EMR systems were developed in North America. Faus and Sujansky (2008) provide a comprehensive review of such systems. In the context of the Canadian Market, OSCAR[2] is an open-source suite of medical applications developed at McMaster University and deployed in multiple clinics in Canada including clinics in British Columbia, Ontario, and Quebec.

This study examines whether the success of open-source software may translate to the medical field. The study looks into the specifics of open-source development in order to decide whether it may resolve some of the problems and challenges that are inhibiting the adoption and the electronic medical record systems. The study will try to answer the following two questions: Will medical workers favor open-source EMR system over proprietary systems? How the development process of an open-source HIT differs from that of proprietary HIT systems?

Finally, from a methodological point of view, open-source information systems present a big opportunity for IS researchers because of the abundance of accessible data. Being open in nature, open-source systems give the opportunity to examine the evolution of the system with a micro perspective by looking at the different development stages. Not only source code of all releases is available, but communication within the community of users and developers is usually archived on the software website. Moreover, because the open-source initiative encourages sharing, the community of

open-source software is also more open and accessible for researchers. In this study I will draw some examples from an ambulatory clinic in Montreal that is adopting the OSCAR system.

[2] www.oscarcanada.org

2. LITERATURE REVIEW AND CONCEPTUAL FRAMEWORK

2 Literature Review and Conceptual Framework

As we have seen in the previous section, the research concerning EMR is very vast and multidisciplinary. To facilitate the discussion of the vast literature and issues of EMR, I will frame my discussion using Swanson and Ramiller's framework of IT innovation. Swanson and Ramiller (2004) separate organizational innovation to four phases: Comprehension, adoption, implementation, and assimilation.

2.1 Comprehension

The comprehension phase is related to the sense-making effort of the organization. In the context of EMRs, the comprehension in a medical unit is recognizing the importance of EMRs, learning about them, and positioning the unit toward adopting them. It is mostly associated with evaluating the potential positive or negative impacts of implementing EMR systems on the outcomes of the medical unit. Accumulating evidence demonstrates the importance of EMR systems in increasing the quality of care and cutting the cost of healthcare.

Improving the Quality of Care

DesRoches et al. (2008) show that in the United States physicians who use electronic health records are happy with their systems and believe that these systems help in improving the quality of care. Yet, only 4% of physicians reported having a fully functional EMR system and 13% have reported having a basic system.

Garrido, Jamieson, Zhou, Wiesenthal, and Liang (2005) studied the impact of EMR systems on the number of visits in ambulatory care clinics. The study found that an EMR system helped in reducing the number of visits that are unnecessary or marginally productive. Specifically, two years after the implementation of EMRs in different sites in the US, age adjusted primary care visit fell by 11% and speciality care visits dropped by 6%. Combined together age adjusted rates of office visits decreased by 9%.

One particular study shows some of the advantages of implementing a computerized physician order entry (CPOE) system (Davidson & Chismar, 2007). The implementation of the system was associated by significant reductions of medication errors, significant increase in the effectiveness of medication uses, an increased compliance to clinical standards, and an improve in clinical administration.

Finally, Tamblyn et al. (2006) studies the implementation of an integrated electronic prescribing and drug management system in primary care clinics. They found that physicians believed that the system would improve the continuity of care, and that they are more willing to adopt it in more complex, fragmented situations.

Cutting Down Healthcare Costs

Hillestad et al. (2005) estimate that, by improving health care efficiency and safety, the widespread

adoption of EMR systems in the US can save more that $81 billion annually. This figure could double if advanced functionalities of EMR systems such as enabled prevention and management of

chronic disease are realized. Walker et al. (2005) report similar findings—the value of fully standardized health care information exchange and interoperability among providers and consumers could be more than $77.8 billion per year. Nonstandardized implementations yield smaller positive returns as well.

Unexpected and Negative Implications

In an ethnographic analysis, Ventres et al. (2006) found that the introduction of an EMR intro practice triggered several cognitive and social changes in the clinical encounter. The study identified 14 factors of how EMRs are used and perceived. The factors are classified into four classes:

- Spatial: impact of the location of the EMR on physician-patient interaction.
- Relational: how physicians and patients perceptions of the EMR affect its use.
- Educational: issues of physicians training and patients understanding of the EMR.
- Structural: institutional and technological factors that affect the use of the EMR.

Finally, HIT systems are not without disadvantages. Software bugs and human errors may propagate and create serious problems especially in large-scale EMRs. Wilson (2009) reports a story that happened in the VA hospitals across the US in 2008. A software problem led to major treatment errors that included patients receiving incorrect doses of drugs and experiencing treatment delays. Another study found that the introduction of a CPOE system resulted in an increase of certain types medication errors (Koppel et al., 2005).

2.2 Adoption

The adoption phase implies a deeper consideration of the IT innovation. This consideration includes weighting the values of the innovation against the challenges presented by the prospective change (Swanson & Ramiller, 2004).

Despite the importance of EMR systems, only 4% of physicians reported having a fully functional EMR system and 13% have reported having a basic system (DesRoches et al., 2008). This rate varies across different medical specialities but it remains low. For example, Kemper, Uren, and Clark (2006) reports that 21.3% of surveyed primary care pediatricians have an EMR system in their clinic. The chance of having such a system increases with the clinic size and cost is the main barrier for nearly all of pediatricians without an EMR.

The situation in hospitals in even worse. In a recent study, Jha et al. (2009) report that only 1.5% of US hospitals have a comprehensive EMR system and only 7.6% have a basic system. Computerized physician order entry (CPOE) systems were implemented in 17% of hospitals. Again, purchasing and maintenance costs are the primary barriers of implementing such systems in hospitals.

Five broad issues must be addressed to promote successful implementation of electronic health records in a small office: financing; interpretability, standardization, and connectivity of clinical information systems; help with redesign of work flow; technical support and training; and help with change management (Baron, Fabens, Schiffman, & Wolf, 2005).

2. LITERATURE REVIEW AND CONCEPTUAL FRAMEWORK

Cost

As we have previously seen, cost is the primary barrier of adopting an EMR system. A recent estimate of the cost of purchasing and implementing a clinic EMR system in the US is $15,000 to $50,000 per physician (Blumenthal & Glaser, 2007). Operating costs may reach 20,000 per year (Miller & West, 2007).

One of President Obama's HITTECH Act goal is to encourage physicians to adopt EMR systems by giving them incentives to implement such systems. The act allocates 19 billion dollars to promote the adoption of EMR, $17 billion of which is allocated to physicians and hospitals (Wilson, 2009)

Coat soars when the system is implemented in a larger organization. The computerized physician order entry (CPOE) project studied in (Davidson & Chismar, 2007) took three years to complete—1992 to 1995—and costed 18 million dollars. The hospital where the system is implemented has 550 beds and over 3,500 employees.

Finally, nationwide implementations require huge investment to be realized. The Report of the Task Force on the Funding of the Health System in Quebec (Castonguay, 2008) cites the example of the Quebec Health Record, a project initiated in 2005 by the Government of Quebec to implement a province wide computerized health record for every patient in Quebec. The initial budged of the project was $562 million, $303 of which will come from Canada Health Infoway and the rest from the Ministry of Health and Social Services in Quebec.

Complexity

EMR systems should be user-friendly because medical workers are not generally tech savvy, nor they have a lot of time to learn a new system. In one case, the transition to EMR system slowed down physicians and resulted in a 10% to 20% reduction in productivity for a period of months or more (Wang et al., 2003). Baron et al. (2005) share their experience in implementing an EMR in their clinic. Staff has to follow two days in-site training before going live with the system. Despite of that everyone in the office become incompetent in doing their job after going live. As a result, the schedule of the clinic was reduced by 50% in the first three days of using the system.

Interoperability

Because patients need to interact with multiple health workers and insurers in multiple health units, patients data should be portable and compatible with the different systems used in different sites. This issue has become very relevant recently with the increase rate of adoption of EMR systems. The solutions differ depending on the country.

Some countries such as UK and Taiwan chose to implement a centralized EMR system to be used nationwide. In North America, however, centralization is not an option because of geography and cultural values. One solution is to store some data centrally and allow access to it by different systems. The Ministry of Health of Quebec has initiated a pilot project to store records of patients in Quebec centrally (Castonguay, 2008). An alternative to centrality is to allow multiple systems by multiple vendors but require these systems to use a common protocol for communication. Health Level Seven

(HL7), a non-for-profit organization accredited by the American National Standard Institute, has been developing standard protocols—that have the same name HL7—for exchang

ing healthcare information in the last two decades (Kush, Helton, Rockhold, & Hardison, 2008). Canada Health Infoway[3] is a Canadian non-for-profit organization that certify EMR systems and invest in their development and use in Canada.

Privacy

So far the previous adoption issues were mainly from the physicians side. However, there are patients concerns that need to be resolved as well. One such issue is privacy. Angst and Agarwal (2006) found that patients concerns of privacy may impede the diffusion of an EMR system that has been demonstrated to reduce medical errors. The privacy issue is not only technical, patients perceptions and education play an important role on bringing this issue as Angst and Agarwal proposed.

2.3 Implementation

The implementation phase is identified by the actions and choices the follow the adoption decision (Swanson & Ramiller, 2004). In their study of the sociological aspect of medical records, Berg and Bowker (1997) argue that the implementation of EMRs is not merely a technical problem of designing and implementing the appropriate interface.

> When it is acknowledged that the medical record is interwoven with the structure of medical work in fundamental ways, and that different medical record systems embody different notions of how work is organized, different modes of configuring patient bodies and so forth, we are in a position better to understand and intervene upon the issues at stake. (Berg & Bowker, 1997, p. 532)

In the context of this study, I will focus on the social issues related to the implementation of EMR systems without discussing the technical issues.

Several studies looked into the cultural issues surrounding the implementation of EMR systems. Kaissi, Kralewski, Dowd, and Heaton (2007) found that the fit between organizational culture—beliefs, values and artifacts—and organizational structure—recurrent patterns of routing interaction and behavior—affected the rate of medication errors when using electronic medical record and drug interaction systems. They conclude that administrators should consider the culture of their organizations before implementing new structures such as EMRs.

Supporting this finding, Baron et al. (2005) highlighted several key decisions they made when implementing an EMR system in their clinic. One key decision was to minimize the IS impact on physician-patient interaction. To achieve this goal, tablet computers with wireless connection were chosen. Another key decision was not to impose any additional requirement on the physicians because of the new system.

> We operate under the assumption that the physician is the most skilled, and most ex pensive, person in the office and should only do what no one other than a physician

could do. (Baron et al., 2005, p. 223)

[3]www.infoway-inforoute.ca

2. LITERATURE REVIEW AND CONCEPTUAL FRAMEWORK

When the system implementation contradicts with cultural values or induces undesirable changes in the workprocess, episodes of resistance occur. Rivard, Lapointe, and Kappos (2009) show that difficulty of implementing an EMR system depends on its effects on the different cultural values in the hospital. They identified four values: quality of care, efficiency of clinical practices, medical dominance, and professional autonomy. Then they classified these values using Martin's three dimensions of a culture: integration, differentiation, and fragmentation (Martin, 1992). Quality of care and clinical practices are integrative values—their interpretation is shared across all medical workers. Medical dominance and professional autonomy are differentiating values—they differ depending on the subgroup (doctors, nurses, technicians, etc. . .). Implementation is facilitated when it is consistent with the four values.

Differentiating values are closely related to the work process and power. Other studies support this finding. Beaudry and Pinsonneault (2005) present a framework that relies on coping to explain users adoption of an IT system. Once the system is implemented, users try to relate the systems to their own objectives and benefits. Users assess the advantages and opportunities that they get from IT and at the same time evaluate the threats and risks that the system poses. Users seek to increase their efficiency and effectiveness, minimize the negative consequences of IT, and restore their emotional stability. Lapointe and Rivard (2005) studied the implementation of three EMR systems in three hospitals. In one case, they found that physicians resistance to the system was triggered by its slowing down their process of carrying out medical procedures. Physicians resistance was passive at the beginning but started to become active when the distribution of power between physicians and nurses was affected—physicians felt the the system is making them do more clerical tasks

Other stories about passive resistance are reported. Weingart et al. (2003) studied the override rate of drug alerts generated at 5 adult primary care practices that uses a common computerized physician order entry (CPOE) system. They found that doctors overrode 91.2% of drug allergy alerts and 89.4% of high-security drug interaction alerts. Similar results were reported by Payne, Nichol, Hoey, and Savarino (2002). At one VA hospital, practitioners overrode 88% and 69% of critical drug interaction and allergy-drug interaction alerts respectively. Tamblyn et al. (2008) reported similar results which suggests that either the alert thresholds in the system are inappropriate or that the doctors feel uncomfortable with following the suggestions of the system.

In their study of a CPOE system, Davidson and Chismar (2007) found that the hospital was careful not to affect the physicians professional autonomy regarding decision-making. Moreover, using the new system was optional, some physicians opted-out from using the system. Instead nurses entered the orders for them into the system. Davidson and Chismar (2007) also found that several customization to the system interface were implemented to match the work process in the hospital.

The role of technology seems different than its role in the famous study of Barley (1986). Barley studied the structural changes that the implementation of identical CT scanners triggered in two similar hospitals. He found that those changes were different in the two hospitals. At one hospital the implementation caused a significant change in the distribution of power in the work process. At another hospital the power relationships between radiologists and technicians did not differ after the implementation of the technology. Contrary to CT scanners that are core medical technologies, EMR systems are a supportive technology. Radiologists *have to* deal with CT scanners whereas doctors can resist EMR systems. In the case of EMR, doctors may choose to resist the system in order to keep their

work process and power status. Therefore, an important

criterium of a successful EMR implementation is to minimize these affects (Baron et al., 2005).

2.4 Assimilation

The assimilation phase is identified by the blending of the IT innovation within the worklife of the organization (Swanson & Ramiller, 2004). From the previous discussion, it is clear that the health-care industry is still far from the assimilation phase. The main challenges today are in the adoption and the implementation of EMR systems. The following section will propose a framework for understanding the adoption and implementation stages of EMR systems.

3 Theoretical Development

3.1 Adoption

As discussed in the previous session, several factors inhibit the adoption of EMR systems. These factors are classified in two groups: primary and secondary factors. Primary factors are the main factors that a medical unit is concerned about when deciding to adopt an EMR system. Cost and complexity are two main factors. Secondary factors on the other hand, affect also the adoption of the system but to a lesser extend than the primary factors. Interoperability and privacy are two secondary factors. In making this difference, I am not suggesting that primary factors are more important than secondary factors. It is just that decision makers at in the medical unit are more concerned of primary factors. Figure 2 shows the different factors affecting the adoption of EMRs.

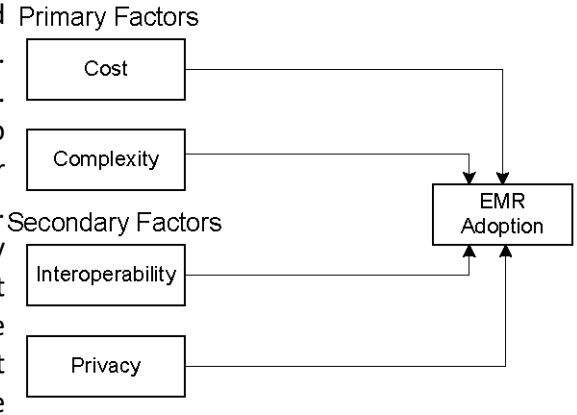


Figure 2: EMR adoption model

Four hypotheses about the adoption of EMR systems are derived from the model:

Hypothesis 1 *The increased cost of purchasing and maintaining of EMR system decreases the adoption of such systems.*

Hypothesis 2 *EMR systems that are complex and not user-friendly will have less adoption rates than simple easy to use systems.*

Hypothesis 3 *EMR systems that are interoperable and support standard storage format and communication protocols will have higher adoption rates than other EMR systems that do not support these features.*

Hypothesis 4 *Privacy of patient data is a concern when adopting EMR systems. Increased support of privacy in EMR systems will increase their adoption rate.*

Adoption of Open-Source EMR

As previously discussed, cost remains the main barriers for adopting EMR systems. EMR software offers a significant cost advantage. In fact, the inial cost of acquiring such systems is almost zero. Users, however, have to pay for support and maintenance of the system. In one of the sites that is implementing OSCAR, the decision to adopt the open-source system stemmed from its reduced initial and maintenance costs. When evaluating the different alternatives, the clinic found that the open-source EMR system will cost fifth of the cost of alternative commercial solutions. This leads to the following hyothesis:

Hypothesis 5 *The reduced cost of open-source EMR systems will increase the likelihood of adopting them compared to proprietary counterparts.*

3.2 Implementation

The implementation phase is more subtle than the adoption phase because it requires *diachronic analysis* (Barley, 1990) therefore I suggest that a process-based model (Markus & Robey, 1988) is more suitable to study this phase in order to trace the unfolding process of the implementation.

The software engineering literature identifies six main stages in software development: Requirement analysis, system design, coding and programming, integrating and testing the system, and finally using and maintaining it (Sommerville, 2000).

Software development models differ according to the emphasis and the order of the different stages. EMR systems are designed to digitize the patient medical records. In this sense, EMR systems have well understood requirements and functionalities. It is argued that for such systems the waterfall and the iterative software development models are very suitable. These models relies of repeating the different stages in order to come to the final release.

Figure 3 depicts the implementation process model of an EMR system. The model is an iterative model that repeats the different stages of implementation in every iteration. In each major phase there developers have the chance of interacting with stakeholders—medical workers—and getting feedback. As discussed before, the main challenges of implementing EMR systems are not technical, they are indeed political and social issues related to the major stakeholders (Berg & Bowker, 1997). In the context of this study, I will focus on the role of feedback which reflects and shapes the interaction between users and developers of the EMR system.

Users Role

System developers seek users feedback during the different stages of the development process—requirement analysis, design, implementation, testing and maintenance. Software users are the

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Figure 3: EMR software implementation process

major stakeholders in any software project. In the system does not satisfy the requirements of the users, they will not use it. Moreover, as discussed previously, if the implementation threatens to change the work relationships or processes at the clinic, group resistance—mainly physicians resistance—occurs. This leads to the following two propositions:

Proposition 1 *Increased feedback from users during the implementation process will increase the quality of the produced EMR software.*

Proposition 2 *Active participation of medical stakeholders in the implementation of an EMR system will reduce the chance of their resistance to the system.*

Users Role in Open-Source Software Development

The development of open-source software relies mainly on the community of voluntary and independent programmers and developers. Users are also involved in the development process by interacting with developers via the software community and by providing feedback. While this is also the case in proprietary software development, the frequency and intensity of feedback differs in the open-source development model.

In the open-source software development model, the source code is available for use and modification by any person or party interested in the project. While this is in theory the case, in practice a limited set of people is interested in developing and modifying the software, and the rest are users who just use it, and request some modifications if necessary, but they do not make an explicit effort to help to or modify the code themselves. Therefore, it is expected that the community of users of an open-source EMR system will be bigger and more varied than the community of a proprietary EMR system. The feedback generated from the community will help in reducing the number of bugs in the software, fastening the development process, and producing the required software.

It is also important to note that users of open-source systems are more active in general than users of proprietary systems. Users of open-source systems do not wait until they are contacted be developers or the organization sponsoring the software. In the contrary, if involved in the software,

users can interact with themselves and developers via the software community using different types of online communication such as mail-lists, online and forums. This active communication positively affect the speed of development and the quality of the produced software.

The previous discussion leads to the following propositions:

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Proposition 3 *Due to a bigger community size and more users interest, feedback intensity and frequency in open-source EMR systems are much larger than those of proprietary EMR systems. This leads to a faster implementation stage and a better quality of the produced software.*

Proposition 4 *Users involvement in the open-source EMR systems relates positively with the quality of the produced software.*

Proposition 5 *Because users of open-source EMR systems are more involved in the implementation process than users of proprietary EMR systems, less resistance to open-source EMR systems is expected to occur during the implementation stage.*

Developers Role

Unless working under a contract, proprietary software vendors cater for the general user in order to create products that suite the largest possible numbers of potential customers. The vendor makes a choice to accept or neglect certain requirements in order to limit cost and satisfy the most prominent customers (Muffatto, 2006). The software has to be released in a specified deadline. Therefore, the feedback on a certain stage cannot be taken into account if the stage is over and development is on the next stage. Such feedback is delayed to the next iteration. Moreover, the feedback source is limited to clients that the vendor deals with. Testing is also done in-house with limited testers and resources.

There is more dynamic in the implementation of an open-source system than a proprietary system. This is because various stakeholders interact and participate in the development process. Contrary to proprietary software development, open-source development is driven by users needs. There are no strict deadlines and the software is released when it is matured. In addition, the community is open for anyone to participate and all opinions, if expressed well, should be considered. Product revision and testing is done faster because of the participation of many community members.

The following propositions about open-source EMR software are proposed:

Proposition 6 *Because of the wealth of feedback generated by the community, developers of open-source EMR software will have a better idea about the true requirements of the software than developers of proprietary EMR software.*

Proposition 7 *The development cycle of open-source EMRs will be longer than that of proprietary EMRs. However, the produce open-source EMR will be more mature.*

Proposition 8 *Compared to proprietary EMR software, open-source EMR systems will come with less bugs and annoyances due to the larger number and variety of testers.*

Having been observing one clinic that is implementing OSCAR, I can partially support the previous propositions about users and developers. OSCAR has a vibrant community of users who interact mainly through the mail-list of the software. OSCAR's evolution is mainly driven by the needs and suggestions of its users. Two doctors in the clinic are active in the user community of Oscar. They provide frequent

feedback, requirements requests and bug fixes to the developers.

4. CONCLUSION

4 Conclusion

Electronic medical record systems hold the promise to profoundly impact quality and cost of care and has the potential to transform care delivery (Faraj, 2008). However, the diffusion of this technology is challenged by many thorny issues. This study investigates the different stages of deploying EMR systems in order to identify the challenges that surround it.

The stages under consideration in this study are the adoption and the implementation stages. It seems that cost is still the major barrier of adopting EMR systems. Other barriers such as complexity, interpretability and privacy exist. During the implementation stage, resistance of medical workers—mainly physicians—is the main challenge that faces EMR systems. Feedback is a key determinant of the quality of produced software and the satisfaction of its users. Group resistance is alleviated when the groups participate in the development process.

The open-source software development model presents many opportunities that may be tapped to deal with the challenges that inhibit the adoption and implementation of EMR systems. Open-source unique characteristics such as free distribution of source-code and community based development may be the solutions for the previously mentioned challenges. More research is required in order to support this hypothesis. The study proposes to carry on longitudinal research across several medical units that are adopting and implementing open-source and proprietary EMR systems.

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